



Airoha IoT SDK for BT Data and Smart Home Open Source Software Guide

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1.0	16 Sep 2025	Initial release.
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1. Overview

This guide provides information about the open-source software bundled in the SDK. It provides information about the open-source software packages and guides the developers in designing, prototyping, and implementing projects in a convenient environment.

As an open-source software package, the information is already available in various sources. This guide serves as an easy reference to module descriptions, including the official website and the hardware or software integration versions. Developers can access the latest source code from the official website and merge or replace the code in the package with a corresponding version. From the footprint statistics results, developers can easily estimate the code size for setting up the flash memory layout. Before commencing your own IoT project or application development, look for example applications and their source code. Finally, the troubleshooting and limitations chapter provides more detailed information on reported issues that may be encountered during the application development.

1.1. Open Source Software Resources for the SDK

The open-source software packages in the SDK are listed in the `<chip_name>_3rd-party_Software_Component_Guide.html` under the `mcu/doc` folder, providing module descriptions, license information, software versions, and links to the official websites.

These resources are intended for the developer's reference and are implemented by the active open-source community, with widely available online resources and forum support. Developers can find the list of online resources for each module in the provided HTML file. Additionally, integrated versions are included, allowing developers to merge hot bug fixes or new features directly from the official release links.

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Table 1 lists the supported modules on different chips. The chips of each product line are shown below.

Table 1. Modules Based on Different Products

Module	BT Data (HID)	BT Data (ESL)	BT Data (Finder)	Smart Home
MCU Boot	✓	✓	✓	✓
Zephyr	✓	✓	✓	✓
Matter				✓
Micro ecc			✓	
LZMA decoder	✓	✓	✓	✓

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Heatshrink		✓		
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The following sections provide a more detailed description of each module.

2. Toolchain

Applicable to 162x only. The Airoha SDK uses the xPack GNU RISC-V Embedded GCC toolchain and links the following two libraries located in the toolchain distribution.

2.1. Newlib

It is a C-standard library implementation intended for use on embedded systems.

License: Refer to COPYING.NEWLIB under `airoha/risc-v/tools/toolchain/xpack-riscv-none-elf-gcc-13.2.0-2/distro-info/licenses/newlib-4.3.0.20230120/`.

Official website: <https://sourceware.org/newlib/>

Version: 4.3.0.20230120

2.2. Libgcc (libgcc.a)

libgcc is a GCC runtime support library, providing compiler helper routines for embedded targets.

License: Refer to COPYING.RUNTIME under `airoha/risc-v/tools/toolchain/xpack-riscv-none-elf-gcc-13.2.0-2/distro-info/licenses/gcc-13.2.0/`.

Official website: <https://gcc.gnu.org/>

Version: GCC 13.2.0

3. MCU Boot

MCUboot is an open-source bootloader for microcontrollers. It is designed to be a secure and flexible bootloader that can be used across a wide range of microcontroller platforms.

3.1. Features

- **Security:** MCUboot provides secure boot capabilities, ensuring that only authenticated and authorized firmware can be executed on the device. It supports cryptographic verification of firmware images using digital signatures.
- **Portability:** MCUboot is designed to be portable across different microcontroller architectures and platforms. It is written in C and can be easily adapted to various hardware environments.
- **Modularity:** The bootloader is modular, allowing developers to enable or disable features based on their specific requirements. This makes it suitable for a wide range of applications, from simple to complex.
- **Firmware Updates:** MCUboot supports firmware update mechanisms, including over-the-air (OTA) updates. It can handle multiple firmware images and provides mechanisms for rollback in case of update failures.
- **Small Footprint:** MCUboot is designed to have a small memory footprint, making it suitable for resource-constrained devices.

3.2. Memory Usage

The expected size is typically less than 64 KB, depending on the feature set.

3.3. Examples

None.

3.4. Customization

AES encryption for secure boot has been added since our platform has HW that supports it.

Customized OTA protocol has been implemented.

3.5. Limitations

None.

3.6. Developer Notes

None.

4. Zephyr

Zephyr is a scalable real-time operating system (RTOS) designed for embedded devices. It offers support for multiple hardware architectures, such as ARM, x86, and RISC-V. Key features include low power consumption management, a modular and extensible architecture, a wide range of device drivers, and built-in security features, like secure boot and access control mechanisms. Zephyr is suitable for IoT applications and edge computing scenarios due to its flexibility and rich ecosystem.

4.1. Features

- Extensive suite of Kernel services: Zephyr offers multi-threading services, interrupt services, memory allocation services, inter-thread synchronization services, inter-thread data passing services, and power management services.
- Multiple Scheduling Algorithms: Zephyr provides cooperative and preemptive scheduling, earliest deadline first (EDF), meta IRQ scheduling, time slicing, and multiple queuing strategies.
- Highly configurable / Modular for flexibility: Allows an application to incorporate only the capabilities it needs, as it needs them, and to specify their quantity and size.
- Cross Architecture: Supports a wide variety of [supported boards](#) with different CPU architectures and developer tools. Contributions have added support for an increasing number of SoCs, platforms, and drivers.
- Memory Protection: Implements configurable architecture-specific stack-overflow protection, kernel object and device driver permission tracking, and thread isolation with thread-level memory protection on x86, ARC, and ARM architectures, user space, and memory domains.
- Compile-time resource definition: Allows system resources to be defined at compile-time, which reduces code size and increases performance for resource-limited systems.
- Optimized Device Driver Model: Provides a consistent device model for configuring the drivers that are part of the platform / system and a consistent model for initializing all the drivers configured in the system and allows the reuse of drivers across platforms that have common devices / IP blocks.
- Devicetree Support: Use of [devicetree](#) to describe hardware. Information from devicetree is used to create the application image.
- Native Networking Stack supporting multiple protocols: Networking support is fully featured and optimized, including LwM2M and BSD sockets compatible support. OpenThread support (on Nordic chipsets) is also provided - a mesh network designed to securely and reliably connect hundreds of products around the home.
- Bluetooth Low Energy 5.0 support: Bluetooth 5.0 compliant (ESR10) and Bluetooth Low Energy Controller support (LE Link Layer). Includes Bluetooth Mesh and a Bluetooth qualification-ready Bluetooth controller.
- Settings on non-volatile storage (NVS): The settings subsystem gives modules a way to store persistent per-device configuration and runtime state. Settings items are stored as key-value pair strings. NVS allows the storage of binary blobs, strings, integers, longs, and any combination of these.

4.2. Memory Usage

Depends on feature set. Zephyr provides [optimizations](#) guide for footprint.

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4.3. Examples

Refer to Zephyr's official [Samples and Demos](#) for the latest detailed information.

4.4. Customization

Refer to Zephyr's official [Configuration System \(Kconfig\)](#) for the latest detailed information.

4.5. Limitations

None.

4.6. Developer Notes

Refer to Zephyr's official [Application Development](#) for the latest detailed information.

4.7. License

The zephyr, along with its integrated third-party modules, is subject to their respective licensing terms. Ensure that you review these modules and their licensing terms to comply with relevant regulations and licensing requirements.

For licensing information regarding Zephyr and its third-party modules, refer to the official documentation: [Zephyr Licensing](#).

5. Matter

Matter (formerly Project CHIP) creates more connections between more objects, simplifying development for manufacturers and increasing compatibility for consumers, guided by the Connectivity Standards Alliance.

5.1. Features

Matter is developed with the following goals and principles in mind:

- **Unifying:** Matter is built with and on top of market-tested, existing technologies.
- **Interoperable:** The specification permits communication between any Matter-certified device, subject to users' permission.
- **Secure:** The specification leverages modern security practices and protocols.
- **User Control:** The end user controls authorization for interaction with devices.
- **Federated:** No single entity serves as a throttle or a single point of failure for the root of trust.
- **Robust:** The set of protocols specifies a complete lifecycle of a device — starting with the seamless out-of-box experience, through operational protocols, to device and system management specifications required for proper function in the presence of change.
- **Low Overhead:** The protocols are practically implementable on low compute-resource devices, such as MCUs.
- **Pervasive:** The protocols are broadly deployable and accessible by leveraging IP and being implementable on low-capability devices.
- **Ecosystem-Flexible:** The protocol is flexible enough to accommodate deployment in ecosystems with differing policies.
- **Easy to Use:** The protocol provides a smooth, cohesive, integrated provisioning and out-of-the-box experience.
- **Open:** The Project's design and technical processes are open and transparent to the general public, including non-members wherever possible.

5.2. Memory Usage

In the project of Airoha Bulb in 202510, the ROM usage of Matter is about 240+ KBytes, and the RAM is 50+ KBytes.

5.3. Examples

Refer to [Matter's official examples](#) for the latest detailed information.

5.4. Customization

None.

5.5. Limitations

None.

5.6. Developer Notes

Refer to [Matter's official handbook](#) for the latest detailed information.

5.7. License

The zephyr, along with its integrated third-party modules, is subject to their respective licensing terms. Ensure that you review these modules and their licensing terms to comply with relevant regulations and licensing requirements.

If your project includes Smart Home features and uses the Matter protocol, note:

- Matter is released under the Apache 2.0 license, but its related third-party modules are subject to their respective licensing terms.
- The Matter project does not consolidate all third-party licenses into one file or web link; relevant licensing documents are provided within each module subdirectory.
- Refer to the LICENSE file or licensing documents in each subdirectory (especially the third_party directory).

6. Micro ECC

A small and fast ECDH and ECDSA implementation for 8-bit, 32-bit, and 64-bit processors.

6.1. Features

- Resistant to known side-channel attacks.
- Written in C, with optional GCC inline assembly for AVR, ARM, and Thumb platforms.
- Support for 8-bit, 32-bit, and 64-bit architectures.
- Small code size.
- No dynamic memory allocation.
- Support for five standard curves: secp160r1, secp192r1, secp224r1, secp256r1, and secp256k1.
- BSD 2-clause license.

6.2. Memory Usage

None.

6.3. Examples

None.

6.4. Customization

None.

6.5. Limitations

None.

6.6. Developer Notes

Refer to <https://github.com/kmackay/micro-ecc>.

7. LZMA Decoder

The LZMA is an algorithm that performs lossless data compression. The LZMA SDK provides a high compression ratio and fast decompression. The Airoha IoT SDK only uses the LZMA decoder algorithm.

7.1. Features

The LZMA decoder provides functions to decompress the compressed data encoded by the LZMA encoder. The LZMA decoder has the following features:

- Small memory requirements: 8 to 32 kB + Dictionary Size.
- Small code size: 2 to 8 kB, depending on speed optimizations.

More information on the LZMA and its features can be found [here](#).

7.2. Memory Usage

Table 2 shows the memory usage.

Table 2. LZMA Decoder Memory Usage

ROM (bytes)	RAM, Static Analysis (in bytes)
6640	32*1024 (Suggested decoding buffer)

7.3. Examples

LZMA decoder module is used to decode FOTA package file in bootloader only. Because OTA binary size is usually too large to have enough buffer for the whole file data, there is a wrapper function `lzma_decode2flash()` provided in `<root>/airoha/subsys/third_parity/lzma_decoder/inc/lzma_decoder_interface.h` header file to decode data block by block, then call HAL flash API to write the output data to the specified address on NOR flash.

More information on the LZMA decoder interface can be found in the bootloader module's source file (`<root>/airoha/mcu/applications/mcuboot/boot/fota/src/bl_fota.c`).

7.4. Customization

None.

7.5. Limitations

None.

7.6. Developer Notes

The LZMA decoder integrated in the current version of the SDK only supports bootloader decoding.

8. Heatshrink

A data compression / decompression library for embedded / real-time systems.

8.1. Features

- Low memory usage (as low as 50 bytes): It is useful for some cases with less than 50 bytes, and useful for many general cases with < 300 bytes.
- Incremental, bounded CPU use: You can chew on input data in arbitrarily tiny bites. This is a useful property in hard real-time environments.
- Can use either static or dynamic memory allocation: The library doesn't impose any constraints on memory management.

8.2. Memory Usage

Table 3 shows the memory usage.

Table 3. Heatshrink Memory Usage

ROM (bytes)	RAM, Static (bytes)
971	388

8.3. Examples

None.

8.4. Customization

None.

8.5. Limitations

None.

8.6. Developer Notes

None.

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